

Victoria Knapp Perez

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Summary

Physics Ph.D. candidate with a strong background in machine learning, reinforcement learning, and data-driven modeling. Experienced in AI methods such as reinforcement learning (OpenAI Gym), deep learning (PyTorch), and high-performance computing, with an emphasis on automated scientific model discovery and symbolic reasoning. Proven record of translating complex theoretical problems into computationally efficient solutions.

Education

- **University of California, Irvine, USA**, Ph.D. Physics (GPA=4.00) (Expected) June 2026
- **University of California, Irvine, USA**, M.S. Physics (GPA=4.00) 2023
- **National Autonomous University of Mexico, Mexico**, B.Sc. Physics (GPA=9.88/10), Cum Laude 2020

Skills

- Programming & Platforms: Python (7+ years), Java (1+ year), Bash scripting (5+ years), LaTeX (9+ years), Slurm (2+ years), CUDA, and High-Performance Computing (HPC) environments.
- Frameworks and Libraries: PyTorch, FastAI, Hugging Face Transformers, OpenAI Gym, scikit-learn, Pandas, NumPy, SymPy, SageMath, spaCy, TorchText, TorchVision.
- Machine Learning & Data Science: Reinforcement learning (PPO), deep learning (CNNs, RNNs, transformers, GNNs), attention and sequence-to-sequence architectures, and fine-tuning of pretrained models such as BERT, GPT, and ResNet.
- Development Tools & Version Control: Git, Jupyter Notebook, Linux shell scripting.
- Soft skills: Strong communication, collaboration, and ability to perform under pressure.
- Certifications: [Coursera Java \(2020\)](#).

Work Experience

University of California: Irvine, USA 2021-Present

Research Assistant & Teaching Assistant

- Developed computational frameworks to automate the construction of physics-based models that predict experimental observables. Combined reinforcement learning (PPO) with high-performance computing tools (Python, Mathematica, Slurm) to efficiently explore large model spaces, optimize parameter sets, and achieve higher predictive accuracy with minimal input complexity.
- Authored multiple peer-reviewed [publications](#) in theoretical and computational physics, including [one](#) integrating AI-driven methods for automated model discovery.
- Delivered instruction and mentorship across 10+ courses from undergraduate to PhD level, developing strong communication and leadership skills while simplifying complex topics for diverse audiences.

National Autonomous University of Mexico: Mexico City, Mexico 2020-2021

Research Assistant & Teaching Assistant

- Applied numerical and symbolic modeling (Python, Mathematica) to simulate and analyze physical systems. Taught and graded advanced undergraduate courses.

Selected projects

AI-assisted Neutrino Flavor Theory Design - University of California: Irvine 2025

- Developed the “Autonomous Model Builder (AMBer)” framework, a reinforcement-learning system built with Stable Baselines and integrated into a custom end-to-end software pipeline to automate physics-model generation, parameter optimization, and validation against experimental data. Enabled large-scale exploration of theoretical spaces and reduced manual analysis time.
- Tech stack: Python, PyTorch, Stable Baselines, OpenAI Gym, reinforcement learning (PPO), data analysis, high-performance computing (Slurm).

Text Reliability Classification with Fine-Tuned LLMs - Erdos Institute ML Project 2025

- Fine-tuned a BERT model for multi-class truthfulness prediction on a Kaggle dataset, benchmarking against a TF-IDF + logistic regression baseline to quantify transformer advantages.
- Tech stack: Python, PyTorch, Hugging Face Transformers, scikit-learn, Pandas, NumPy.

Selected publications and awards

- J. B. Baretz, M. Fieg, V. Ganesh, A. Ghosh, V. Knapp-Perez, J. Rudolph and D. Whiteson, “Towards AI-assisted Neutrino Flavor Theory Design”, [\[arXiv:2506.08080 \[hep-ph\]\]](#). 2025
- Latino Excellence and Achievement Award (LEAD), honored for research excellence and leadership of Hispanic/Latinx students at UCI. 2024
- UC President’s Lindau Nobel Laureate Meetings Fellowship, selected to engage with Nobel Laureates and international scientists. 2024
- UCI Department of Physics and Astronomy Outstanding Graduate Student Award, recognizing top first-year graduate student. 2022